

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/text1"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="50dp"
        android:layout_weight="1"
        android:background="#E1ACAC"
        android:gravity="center"
        android:text="Here's your Screen"
        android:textColor="#040000"
        android:textSize="25sp"
        android:textStyle="bold"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

    <EditText
        android:id="@+id/uname"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_weight="0"
        android:ems="10"
        android:hint="Please Enter Your Name"
        android:inputType="textPersonName"
        android:text=""
        android:textColorHint="#1B1818" />

</EditText>
```

```

        android:id="@+id/phone"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Please Enter Your Phone number"
        android:textColorHint="#1B1818"/>
    <Button
        android:id="@+id/btn1"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:background="#DFAE9E"
        android:shadowColor="#B67171"
        android:text="Show my Name"
        android:textColorLink="#CF9E9E"
        app:rippleColor="#A65A5A"
        android:textSize="20sp"
        android:layout_marginBottom="50dp"
    "
    />

</LinearLayout>

```

Java code

```

package com.example.editttext;

import android.os.Bundle;
import android.provider.ContactsContract;
import android.util.Log;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;

```

```

import androidx.core.view.WindowInsetsCompat;

import java.text.BreakIterator;

public class MainActivity extends AppCompatActivity {
    private TextView text1;
    private Button btn1;
    private EditText uname, phone;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        text1 = (TextView) findViewById(R.id.text1);
        btn1 = (Button) findViewById(R.id.btn1);
        uname = (EditText) findViewById(R.id.uname);
        phone = (EditText) findViewById(R.id.phone);

        btn1.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {

                String userName = uname.getText().toString();
                String uphone = phone.getText().toString();

                text1.setText("Your name is:" + userName +
                    ",\n" + "Your Phone Number is: " + uphone);

            }
        });
    }
}

```

Textview app

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/textview"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:layout_margin="0dp"
    android:layout_marginLeft="0dp"
    android:layout_marginTop="0dp"
    android:layout_marginRight="0dp"
    android:layout_marginBottom="0dp"
    android:background="@color/white"
    android:padding="0dp"
    android:paddingLeft="0dp"
    android:orientation="vertical"

    android:theme="@style/MaterialAlertDialog.Material3.Title.Panel"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/text1"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:background="#DFD5D5"
        android:gravity="left"
        android:padding="10dp"
```

```

        android:paddingLeft="0dp"
        android:text="Parul University"
        android:textSize="30sp"
        android:textStyle="bold" />

<TextView
    android:id="@+id/text2"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_below="@id/text1"
    android:layout_weight="1"
    android:background="#E88989"
    android:paddingLeft="20dp"
    android:paddingTop="20dp"
    android:text="TextView Program"
    android:textSize="15sp" />

<TextView
    android:id="@+id/text3"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_below="@id/text2"
    android:height="50dp"
    android:background="#DCCECE"
    android:gravity="center"
    android:text="This is an TextView Program"
    android:textSize="20sp" />
</LinearLayout>

```

Mutltielement

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<LinearLayout
xmlns:android="http://schemas.android.com
/apk/res/android"

xmlns:app="http://schemas.android.com/apk
/res-auto"

xmlns:tools="http://schemas.android.com/t
ools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity"
    android:orientation="vertical">

    <CheckBox
        android:id="@+id/chel"

        android:layout_width="match_parent"

        android:layout_height="wrap_content"
        android:gravity="center"
        android:text="VolleyBall"
        android:textStyle="bold"

        tools:layout_editor_absoluteX="36dp"
```

```
tools:layout_editor_absoluteY="17dp" />
```

```
<CheckBox  
    android:id="@+id/che2"
```

```
    android:layout_width="match_parent"
```

```
    android:layout_height="wrap_content"  
    android:gravity="center"  
    android:text="Tennis"  
    android:textStyle="bold" />
```

```
<CheckBox  
    android:id="@+id/che3"
```

```
    android:layout_width="match_parent"
```

```
    android:layout_height="wrap_content"  
    android:gravity="center"  
    android:text="BasketBall"  
    android:textStyle="bold" />
```

```
<CheckBox  
    android:id="@+id/che4"
```

```
android:layout_width="match_parent"
```

```
android:layout_height="wrap_content"
```

```
    android:gravity="center"
```

```
    android:text="Hockey"
```

```
    android:textStyle="bold" />
```

```
<CheckBox
```

```
    android:id="@+id/ch5"
```

```
android:layout_width="match_parent"
```

```
android:layout_height="wrap_content"
```

```
    android:gravity="center"
```

```
    android:text="Cricket"
```

```
    android:textStyle="bold" />
```

```
<SeekBar
```

```
android:layout_width="match_parent"
```

```
    android:layout_height="32dp"
```

```
    android:layout_marginTop="18dp"
```

```
    android:id="@+id/simpleSeekBar"
```

```
    android:indeterminate="false"
```

```
    android:max="200"
```

```
    android:min="0"
```

```
        android:progress="60"
        android:layout_marginLeft="18dp"

    android:layout_marginRight="18dp"/>
    <RatingBar

    android:layout_width="wrap_content"

    android:layout_height="wrap_content"
        android:numStars="5"
        android:layout_gravity="center"
        android:id="@+id/rtn1"
        android:layout_marginTop="18dp"/>
    <RadioGroup

    android:layout_width="match_parent"

    android:layout_height="wrap_content"
        android:id="@+id/rdl1">
        <RadioButton

    android:layout_width="match_parent"

    android:layout_height="wrap_content"
        android:text="Mobile
development"
```

```
        android:id="@+id/radio1"
        android:textStyle="bold"
        android:textSize="25sp"/>
    <RadioButton

    android:layout_width="match_parent"

    android:layout_height="wrap_content"
        android:text="Web development"
        android:id="@+id/radio2"
        android:textStyle="bold"
        android:textSize="25sp"/>
    <RadioButton

    android:layout_width="match_parent"

    android:layout_height="wrap_content"
        android:text="Specilization"
        android:id="@+id/radio3"
        android:textStyle="bold"
        android:textSize="25sp"/>

</RadioGroup>
<Button

    android:layout_width="match_parent"
```

```
        android:layout_height="wrap_content"
        android:layout_marginLeft="30dp"
        android:layout_marginRight="30dp"
        android:layout_marginTop="40dp"
        android:text="click me "
        android:id="@+id/buttons"
    />
</LinearLayout>
```

```
package com.example.multielements;
```

```
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.CheckBox;
import android.widget.RadioButton;
import android.widget.RadioGroup;
import android.widget.RatingBar;
import android.widget.SeekBar;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import
androidx.appcompat.app.AppCompatActivity;
```

```
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import
androidx.core.view.WindowInsetsCompat;

public class MainActivity extends
AppCompatActivity {
    public CheckBox ch1, ch2, ch3, ch4,
ch5;
    public SeekBar seekBar;
    public RatingBar ratingBar;
    public RadioGroup radioGroup; //,
    public RadioButton rb1;
    public Button btn;
    Button submitButton;
    SeekBar simpleSeekBar;

    @Override
    protected void onCreate(Bundle
savedInstanceState) {

super.onCreate(savedInstanceState);

setContentView(R.layout.activity_main);
    //seekbaar
```

```
simpleSeekBar=(SeekBar) findViewById(R.id.  
simpleSeekBar);
```

```
ch1 = findViewById(R.id.che1);  
ch2 = findViewById(R.id.che2);  
ch3 = findViewById(R.id.che3);  
ch4 = findViewById(R.id.che4);  
ch5 = findViewById(R.id.ch5);  
btn = findViewById(R.id.buttons);
```

```
simpleSeekBar.setOnSeekBarChangeListener(  
new SeekBar.OnSeekBarChangeListener() {  
    int progressChangedValue = 0;  
  
    public void  
onProgressChanged(SearchBar seekBar, int  
progress, boolean fromUser) {  
        progressChangedValue =  
progress;  
    }  
  
    public void  
onStartTrackingTouch(SearchBar seekBar) {
```

```
                                // TODO Auto-generated
method stub
    }

    public void
onStopTrackingTouch(SeekBar seekBar) {

Toast.makeText(MainActivity.this, "Seek
bar progress is :" +
progressChangedValue,

Toast.LENGTH_SHORT).show();
    }
});

    btn.setOnClickListener(new
View.OnClickListener() {
        public void onClick(View v) {
            check();
        }
    });

}

public void check() {
```

```

        String msg = "";
        if (ch1.isChecked())
            msg = msg + " " +
"Volleyball";
        if (ch2.isChecked())
            msg = msg + " " +
"BasketBall";
        if (ch3.isChecked())
            msg = msg + " " + "Tennis";
        if (ch4.isChecked())
            msg = msg + " " + "Hockey";
        if (ch5.isChecked())
            msg = msg + " " + "Cricket";

        Toast.makeText(this, "You have
Selected " + msg,
Toast.LENGTH_SHORT).show();

    }
}

```

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com
/apk/res/android"

xmlns:app="http://schemas.android.com/apk
/res-auto"

xmlns:tools="http://schemas.android.com/t
ools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <Button

    android:layout_width="wrap_content"

    android:layout_height="wrap_content"
        android:text="submit"
        android:id="@+id/button"
```

```
app:layout_constraintBottom_toBottomOf="parent"
```

```
app:layout_constraintLeft_toLeftOf="parent"
```

```
app:layout_constraintRight_toRightOf="parent"
```

```
app:layout_constraintTop_toTopOf="parent"
```

```
app:layout_constraintVertical_bias="0.615" />
```

```
<RatingBar
```

```
    android:id="@+id/ratingBar"
```

```
    android:layout_width="wrap_content"
```

```
    android:layout_height="wrap_content"
```

```
        android:layout_marginLeft="72dp"
```

```
        android:layout_marginTop="60dp"
```

```
app:layout_constraintStart_toStartOf="parent"
```

```
app:layout_constraintTop_toTopOf="parent"  
/>
```

```
</androidx.constraintlayout.widget.ConstraintLayout>
```

```
package com.example.ratingbar;
```

```
import android.os.Bundle;  
import android.view.View;  
import android.widget.Button;  
import android.widget.RatingBar;  
import android.widget.Toast;
```

```
import androidx.activity.EdgeToEdge;  
import  
androidx.appcompat.app.AppCompatActivity;  
import androidx.core.graphics.Insets;  
import androidx.core.view.ViewCompat;  
import  
androidx.core.view.WindowInsetsCompat;
```

```
public class MainActivity extends
AppCompatActivity {
    RatingBar ratingbar;
    Button button;
    @Override
    protected void onCreate(Bundle
savedInstanceState) {

    super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);

    setContentView(R.layout.activity_main);
        addListenerOnButtonClick();

    ViewCompat.setOnApplyWindowInsetsListener
    (findViewById(R.id.main), (v, insets) ->
    {
        Insets systemBars =
insets.getInsets(WindowInsetsCompat.Type.
systemBars());
        v.setPadding(systemBars.left,
systemBars.top, systemBars.right,
systemBars.bottom);
        return insets;
    });
}
```

```

        private void
addListenerOnButtonClick() {

ratingbar=(RatingBar) findViewById(R.id.ratingBar);

button=(Button) findViewById(R.id.button);
        button.setOnClickListener(new
View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String
rating=String.valueOf(ratingbar.getRating
());

Toast.makeText(getApplicationContext(),
rating, Toast.LENGTH_LONG).show();
            }
        });
    }
}

```

Alert Dialog prog

Xml code:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com
/apk/res/android"

xmlns:app="http://schemas.android.com/apk
/res-auto"

xmlns:tools="http://schemas.android.com/t
ools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <Button

    android:layout_width="wrap_content"

    android:layout_height="wrap_content"
        android:id="@+id/button"
        android:text="Close app"
```

```
        android:textSize="40sp"

app:layout_constraintBottom_toBottomOf="parent"

app:layout_constraintLeft_toLeftOf="parent"

app:layout_constraintRight_toRightOf="parent"

app:layout_constraintTop_toTopOf="parent"
/>

</androidx.constraintlayout.widget.ConstraintLayout>
```

String.xml code:

```
<resources>
    <string
name="app_name">AlertDialog</string>
```

```
<string name="dialog_message">Welcome  
to Alert Dialog</string>  
    <string name="dialog_title">Parul  
Alert Dialog</string>  
</resources>
```

Java Code :

```
package com.example.alertdialog;  
  
import android.content.DialogInterface;  
import android.os.Bundle;  
import android.view.View;  
import android.widget.Button;  
import android.widget.Toast;  
  
import androidx.activity.EdgeToEdge;  
import  
androidx.appcompat.app.AlertDialog;  
import  
androidx.appcompat.app.AppCompatActivity;  
import androidx.core.graphics.Insets;
```

```
import androidx.core.view.ViewCompat;
import
androidx.core.view.WindowInsetsCompat;

public class MainActivity extends
AppCompatActivity {
    Button closeButton;
    AlertDialog.Builder builder;
    @Override
    protected void onCreate(Bundle
savedInstanceState) {

    super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);

    setContentView(R.layout.activity_main);
        closeButton = (Button)
findViewById(R.id.button);
        builder = new
AlertDialog.Builder(this);
        closeButton.setOnClickListener(new
View.OnClickListener() {
            @Override
            public void onClick(View v) {
                //Setting message manually
                and performing action on button click
            }
        })
    }
}
```

```
        builder.setMessage("Do you  
want to close this application ?")  
  
        .setCancelable(false)  
  
        .setPositiveButton("Yes", new  
DialogInterface.OnClickListener() {  
            public void  
onClick(DialogInterface dialog, int id) {  
                finish();  
            }  
        })  
  
        Toast.makeText(getApplicationContext(), "y  
ou choose yes action for alertbox",  
        Toast.LENGTH_SHORT).show();  
    }  
})  
  
        .setNegativeButton("No", new  
DialogInterface.OnClickListener() {  
            public void  
onClick(DialogInterface dialog, int id) {  
                // Action  
for 'NO' Button  
  
                dialog.cancel();  
            }  
        })  
    }  
}
```

```

Toast.makeText(getApplicationContext(),"you
choose no action for alertbox",

Toast.LENGTH_SHORT).show();

    }

    });

    //Creating dialog box
    AlertDialog alert =
builder.create();
    //Setting the title
manually

alert.setTitle("AlertDialogExample");
    alert.show();

    }

    });

}

}

```

Autocpomp;letetextview

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com
/apk/res/android"

xmlns:app="http://schemas.android.com/apk
/res-auto"

xmlns:tools="http://schemas.android.com/t
ools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView

    android:layout_width="wrap_content"

    android:layout_height="wrap_content"
        android:text="What is your
favourite programming language?"
        android:textSize="20dp"
```

```
app:layout_constraintBottom_toBottomOf="parent"
```

```
app:layout_constraintHorizontal_bias="0.495"
```

```
app:layout_constraintLeft_toLeftOf="parent"
```

```
app:layout_constraintRight_toRightOf="parent"
```

```
app:layout_constraintTop_toTopOf="parent"
```

```
app:layout_constraintVertical_bias="0.338" />
```

```
<AutoCompleteTextView
```

```
android:id="@+id/autoCompleteTextView"
    android:layout_width="200dp"
```

```
android:layout_height="wrap_content"
    android:layout_marginStart="112dp"
    android:layout_marginTop="376dp"
```

```
        android:text=""

app:layout_constraintStart_toStartOf="parent"

app:layout_constraintTop_toTopOf="parent"
/>

</androidx.constraintlayout.widget.ConstraintLayout>
```

Java code

```
package com.example.autocomplete;

import android.graphics.Color;
import android.os.Bundle;
import android.widget.ArrayAdapter;
import
android.widget.AutoCompleteTextView;

import androidx.activity.EdgeToEdge;
import
androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
```

```
import androidx.core.view.ViewCompat;
import
androidx.core.view.WindowInsetsCompat;

public class MainActivity extends
AppCompatActivity {
    String[] language
={ "C", "C++", "Java", ".NET", "iPhone", "Andro
id", "ASP.NET", "PHP" };
    @Override
    protected void onCreate(Bundle
savedInstanceState) {

super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);

setContentView(R.layout.activity_main);
        //Creating the instance of
ArrayAdapter containing list of language
names

        ArrayAdapter<String> adapter = new
ArrayAdapter<String>

(this, android.R.layout.select_dialog_item
, language);
```

*//Getting the instance of
AutoCompleteTextView*

```
AutoCompleteTextView actv =  
(AutoCompleteTextView)findViewById(R.id.a  
utoCompleteTextView) ;  
    actv.setThreshold(1) ;//will start  
working from first character  
    actv.setAdapter(adapter) ;//setting  
the adapter data into the  
AutoCompleteTextView  
    actv.setTextColor(Color.RED) ;  
    }  
}
```

Toggle Button

```
<?xml version="1.0" encoding="utf-8"?>  
<androidx.constraintlayout.widget.ConstraintLa  
intLayout  
xmlns:android="http://schemas.android.com  
/apk/res/android"
```

```
xmlns:app="http://schemas.android.com/apk/res-auto"
```

```
xmlns:tools="http://schemas.android.com/tools"
```

```
    android:id="@+id/main"  
    android:layout_width="match_parent"  
    android:layout_height="match_parent"  
    tools:context=".MainActivity">
```

```
    <ToggleButton  
        android:id="@+id/toggleButton"
```

```
        android:layout_width="wrap_content"
```

```
        android:layout_height="wrap_content"  
        android:layout_marginLeft="8dp"  
        android:layout_marginTop="80dp"  
        android:text="ToggleButton"  
        android:textOff="Off"  
        android:textOn="On"
```

```
    app:layout_constraintEnd_toStartOf="@+id/toggleButton2"
```

```
app:layout_constraintStart_toStartOf="parent"
```

```
app:layout_constraintTop_toTopOf="parent"  
/>
```

```
    <ToggleButton  
        android:id="@+id/toggleButton2"  
  
        android:layout_width="wrap_content"  
  
        android:layout_height="wrap_content"  
        android:layout_marginRight="60dp"  
        android:layout_marginTop="80dp"  
        android:text="ToggleButton"  
        android:textOff="Off"  
        android:textOn="On"
```

```
app:layout_constraintEnd_toEndOf="parent"
```

```
app:layout_constraintTop_toTopOf="parent"  
/>
```

```
    <Button  
        android:id="@+id/button"
```

```
android:layout_width="wrap_content"

android:layout_height="wrap_content"

android:layout_marginBottom="144dp"
    android:layout_marginLeft="148dp"
    android:text="Submit"

app:layout_constraintBottom_toBottomOf="parent"

app:layout_constraintStart_toStartOf="parent" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

Java code

```
package com.example.toggle;

import android.os.Bundle;
```

```
import android.view.View;
import android.widget.Button;
import android.widget.Toast;
import android.widget.ToggleButton;

import androidx.activity.EdgeToEdge;
import
androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import
androidx.core.view.WindowInsetsCompat;

public class MainActivity extends
AppCompatActivity {
    private ToggleButton toggleButton1,
toggleButton2;
    private Button buttonSubmit;
    @Override
    protected void onCreate(Bundle
savedInstanceState) {

super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);

setContentView(R.layout.activity_main);
```

```

        addListenerOnButtonClick();
    }

    public void
addListenerOnButtonClick() {
        //Getting the ToggleButton and
        Button instance from the layout xml file

        toggleButton1=(ToggleButton) findViewById(
R.id.toggleButton);

        toggleButton2=(ToggleButton) findViewById(
R.id.toggleButton2);

        buttonSubmit=(Button) findViewById(R.id.bu
        tton);

        //Performing action on button
        click

        buttonSubmit.setOnClickListener(new
View.OnClickListener() {

            @Override
            public void onClick(View view)
{

```

```

        StringBuilder result = new
StringBuilder();

result.append("ToggleButton1 :
") .append(toggleButton1.getText());

result.append("\nToggleButton2 :
") .append(toggleButton2.getText());
        //Displaying the message
in toast

Toast.makeText(getApplicationContext(),
result.toString(),Toast.LENGTH_LONG).show
();

    }

    });

}
}

```

Datepicker

```
<?xml version="1.0" encoding="utf-8"?>
    <RelativeLayout
xmlns:android="http://schemas.android.com
/apk/res/android"

xmlns:app="http://schemas.android.com/apk
/res-auto"

xmlns:tools="http://schemas.android.com/t
ools"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        tools:context=".MainActivity">

        <TextView
            android:id="@+id/textView1"

            android:layout_width="wrap_content"

            android:layout_height="wrap_content"

            android:layout_above="@+id/button1"

            android:layout_alignParentLeft="true"

            android:layout_alignParentStart="true"
```

```
android:layout_marginBottom="102dp"  
    android:layout_marginLeft="30dp"  
    android:layout_marginStart="30dp"  
    android:text="" />
```

```
<Button  
    android:id="@+id/button1"
```

```
    android:layout_width="wrap_content"
```

```
    android:layout_height="wrap_content"
```

```
    android:layout_alignParentBottom="true"
```

```
    android:layout_centerHorizontal="true"  
    android:layout_marginBottom="20dp"  
    android:text="Change Date" />
```

```
<DatePicker  
    android:id="@+id/datePicker"
```

```
    android:layout_width="wrap_content"
```

```
    android:layout_height="wrap_content"
```

```
    android:layout_above="@+id/textView1"

    android:layout_centerHorizontal="true"
        android:layout_marginBottom="36dp"
/>

</RelativeLayout>
```

```
package com.example.datepicker;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.DatePicker;
import android.widget.TextView;

import androidx.activity.EdgeToEdge;
import
androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
```

```
import
androidx.core.view.WindowInsetsCompat;

public class MainActivity extends
AppCompatActivity {
    DatePicker picker;
    Button displayDate;
    TextView textView1;

    @Override
    protected void onCreate(Bundle
savedInstanceState) {

super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);

setContentView(R.layout.activity_main);
textView1=(TextView) findViewById(R.id.textView1);

picker=(DatePicker) findViewById(R.id.date
Picker);

displayDate=(Button) findViewById(R.id.button1);
```

```
        textView1.setText("Current Date:
"+getCurrentDate());

        displayDate.setOnClickListener(new
View.OnClickListener() {
            @Override
            public void onClick(View view)
            {

                textView1.setText("Change
Date: "+getCurrentDate());
            }

        });

    }

    public String getCurrentDate() {
        StringBuilder builder=new
StringBuilder();
        builder.append((picker.getMonth()
+ 1)+" /");//month is 0 based

        builder.append(picker.getDayOfMonth()+" /"
);

        builder.append(picker.getYear());
        return builder.toString();
    }
}
```

```
    }  
}  
Time Picker
```

```
<?xml version="1.0" encoding="utf-8"?>  
<RelativeLayout  
    xmlns:android="http://schemas.android.com  
        /apk/res/android"  
  
    xmlns:app="http://schemas.android.com/apk  
        /res-auto"  
  
    xmlns:tools="http://schemas.android.com/t  
        ools"  
        android:layout_width="match_parent"  
        android:layout_height="match_parent"  
        tools:context=".MainActivity">  
    <TextView  
        android:id="@+id/textView1"  
  
        android:layout_width="wrap_content"  
  
        android:layout_height="wrap_content"
```

```
android:layout_above="@+id/button1"

android:layout_alignParentLeft="true"

android:layout_alignParentStart="true"

android:layout_marginBottom="102dp"
    android:layout_marginLeft="30dp"
    android:layout_marginStart="30dp"
    android:text="" />

<Button
    android:id="@+id/button1"

    android:layout_width="wrap_content"

    android:layout_height="wrap_content"

    android:layout_alignParentBottom="true"

    android:layout_centerHorizontal="true"
        android:layout_marginBottom="20dp"
        android:text="Change Time" />

<TimePicker
```

```
        android:id="@+id/timePicker"

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:layout_above="@+id/textView1"

        android:layout_centerHorizontal="true"
        android:layout_marginBottom="36dp"
    />
</RelativeLayout>
```

Java code

```
package com.example.timepicker;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;
```

```
import android.widget.TimePicker;

import androidx.activity.EdgeToEdge;
import
androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import
androidx.core.view.WindowInsetsCompat;

public class MainActivity extends
AppCompatActivity {
    TextView textView1;
    TimePicker timepicker;
    Button changetime;

    @Override
    protected void onCreate(Bundle
savedInstanceState) {

super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);

setContentView(R.layout.activity_main);
        textView1 = (TextView)
findViewById(R.id.textView1);
```

```
        timepicker = (TimePicker)
findViewById(R.id.timePicker); // Uncomment
the below line of code for 24 hour view
        timepicker.setIs24HourView(true);
        changetime = (Button)
findViewById(R.id.button1);
```

```
textView1.setText(getCurrentTime());
```

```
        changetime.setOnClickListener(new
View.OnClickListener() {
            @Override
            public void onClick(View view)
{
```

```
textView1.setText(getCurrentTime());
        }
    });
```

```
}
```

```
    public String getCurrentTime() {
        String currentTime="Current Time:
"+timepicker.getCurrentHour()+":"+timepic
ker.getCurrentMinute();
```

```
        return currentTime;
    }

}
```

Listview

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res
/android"

xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    tools:context=".MainActivity" >

    <ListView
        android:id="@+id/mobile_list"
```

```
        android:layout_width="match_parent"
        android:layout_height="wrap_content" >
    </ListView>

</LinearLayout>
```

Then create another xml file for the listing view name is : acitivity_listview.xml

Acitivity_listview.xml

```
<?xml version="1.0" encoding="utf-8"?>
<TextView
xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/label"
    android:layout_width="fill_parent"
    android:layout_height="fill_parent"
    android:padding="10dip"
    android:textSize="16dip"
    android:textStyle="bold" >
</TextView>
```

JAVA CODE

```
package com.example.listview;
```

```

import android.os.Bundle;
import android.view.View;
import android.widget.AdapterView;
import android.widget.ArrayAdapter;
import android.widget.ListView;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import
androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import
androidx.core.view.WindowInsetsCompat;

    public class MainActivity extends
AppCompatActivity {
        String[] mobileArray =
{"Android", "IPhone", "WindowsMobile", "Blackberry"
,
        "WebOS", "Ubuntu", "Windows7", "Max
OS X"};

        @Override
        protected void onCreate(Bundle
savedInstanceState) {
            super.onCreate(savedInstanceState);
            EdgeToEdge.enable(this);

setContentView(R.layout.activity_main);

```

```

        ArrayAdapter adapter = new
ArrayAdapter<String>(this,
                    R.layout.activity_listview,
mobileArray) ;

```

```

        ListView listView = (ListView)
findViewById(R.id.mobile_list) ;
        listView.setAdapter(adapter) ;
        listView.setOnItemClickListener(new
AdapterView.OnItemClickListener() {
            @Override
                public void
onItemClick(AdapterView<?> parent, View view,
int position, long id) {
                    String data=
adapter.getItem(position).toString() ;
//String
v=adapter.getItem(position) ;

Toast.makeText(getApplicationContext(),data,Toas
t.LENGTH_SHORT).show() ;
            }
        });
    }
}

```

-----Notification

```
<?xml                                version="1.0"
encoding="utf-8" ?>
    <androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"

xmlns:app="http://schemas.android.com/apk/res-auto"

xmlns:tools="http://schemas.android.com/tools"
        android:id="@+id/main"

        android:layout_width="match_parent"

        android:layout_height="match_parent"
        tools:context=".MainActivity">

        <TextView

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"
            android:text="Hello World!"
```

```
app:layout_constraintBottom_toBottomOf="parent"
```

```
app:layout_constraintEnd_toEndOf="parent"
```

```
app:layout_constraintStart_toStartOf="parent"
```

```
app:layout_constraintTop_toTopOf="parent" />
```

```
</androidx.constraintlayout.widget.ConstraintLayout>
```

java code :

```
package com.example.notification;
```

```
import android.app.Notification;
```

```
import
```

```
android.app.NotificationChannel;
```

```
import
android.app.NotificationManager;
import android.graphics.Bitmap;
import
android.graphics.drawable.BitmapDrawable
;

import
android.graphics.drawable.Drawable;
import android.os.Bundle;

import androidx.activity.EdgeToEdge;
import
androidx.appcompat.app.AppCompatActivity
;

import
androidx.core.content.res.ResourcesCompa
t;

import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import
androidx.core.view.WindowInsetsCompat;

public class MainActivity extends
AppCompatActivity {
    private static final String
CHANNEL_ID= "My Channel";
```

```

        private static final int
NOTIFICATION_ID= 100;
        @Override
        protected void onCreate(Bundle
savedInstanceState) {

super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);

setContentView(R.layout.activity_main);
                Drawable drawable =
ResourcesCompat.getDrawable(getResources
(), R.drawable.icon, null);
                BitmapDrawable bitmapDrawable
= (BitmapDrawable) drawable;
                Bitmap largeIcon =
bitmapDrawable.getBitmap();
                NotificationManager nm =
(NotificationManager)
getSystemService(NOTIFICATION_SERVICE);
                Notification notification;

                                                                if
(android.os.Build.VERSION.SDK_INT >=
android.os.Build.VERSION_CODES.O) {
                        notification = new
Notification.Builder(this)

```

```
.setLargeIcon (largeIcon)
```

```
.setSmallIcon (R.drawable.icon)
```

```
.setContentText ("Hi, salary of INR  
89,657.00 is credited to HDFC Bank A/c  
XX6567. Aval Bal: INR 10,00,00,000.00")
```

```
.setSubText ("Congratulation")
```

```
.setChannelId (CHANNEL_ID)  
                .build() ;
```

```
nm.createNotificationChannel (new  
NotificationChannel (CHANNEL_ID, "New  
Channel",  
NotificationManager.IMPORTANCE_HIGH) ) ;  
    } else {  
        notification = new  
Notification.Builder (this)
```

```
.setLargeIcon (largeIcon)
```

```
.setSmallIcon(R.drawable.icon)
```

```
.setTextContent("Your Salary got  
credited. Your Current Balance is  
10,00,00,000")
```

```
.setSubText("Congratulation")
```

```
.build();
```

}

```
nm.notify(NOTIFICATION_ID,  
notification);
```

}

}

Broadcast Recievrr

```
<?xml version="1.0" encoding="utf-8" ?>
```

```
<androidx.constraintlayout.widget.Con  
straintLayout  
xmlns:android="http://schemas.android.co  
m/apk/res/android"
```

```
xmlns:app="http://schemas.android.com/ap  
k/res-auto"
```

```
xmlns:tools="http://schemas.android.com/  
tools"
```

```
    android:id="@+id/main"
```

```
    android:layout_width="match_parent"
```

```
    android:layout_height="match_parent"  
    tools:context=".MainActivity">
```

```
    <TextView
```

```
        android:id="@+id/tv"
```

```
    android:layout_width="wrap_content"
```

```
    android:layout_height="wrap_content"
```

```
        android:text="Broadcast  
Recievre"
```

```
app:layout_constraintBottom_toBottomOf="parent"
```

```
app:layout_constraintEnd_toEndOf="parent"
```

```
app:layout_constraintStart_toStartOf="parent"
```

```
app:layout_constraintTop_toTopOf="parent" />
```

```
</androidx.constraintlayout.widget.ConstraintLayout>
```

```
MainActivity.java
```

```
package  
com.example.broadcastreceiver;
```

```
import android.content.Intent;
import android.content.IntentFilter;
import android.os.Bundle;
import android.widget.TextView;

import androidx.activity.EdgeToEdge;
import
androidx.appcompat.app.AppCompatActivity
;

import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import
androidx.core.view.WindowInsetsCompat;

public class MainActivity extends
AppCompatActivity {
    TextView tv;
    BroadcastReceiver batteryReceiver;
    @Override
        protected void onCreate(Bundle
savedInstanceState) {

super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);

setContentView(R.layout.activity_main);
```

```
        tv= findViewById(R.id.tv);
        batteryReceiver = new
BatteryReceiver(tv);

registerReceiver(batteryReceiver, new
IntentFilter(Intent.ACTION_BATTERY_CHANG
ED));
    }

@Override
protected void onStop() {
    super.onStop();
}

unregisterReceiver(batteryReceiver);
}
}
```

BatteryReciver.java

```
package
com.example.broadcastreceiver;
```

```
import
android.content.BroadcastReceiver;
import android.content.Context;
import android.content.Intent;
import android.widget.TextView;

public class BatteryReceiver extends
BroadcastReceiver {
    private TextView tv;

    public BatteryReceiver(TextView
tv) {
        this.tv = tv;
    }

    @Override
    public void onReceive(Context
context, Intent intent) {
        int level =
intent.getIntExtra("level", 0);
        tv.setText("Battery Level: " +
level + "%");
    }
}
```

Shared preference

Activity_main.xml

```
<?xml version="1.0"
encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas
.android.com/apk/res/androi
d"

xmlns:app="http://schemas.an
droid.com/apk/res-auto"

xmlns:tools="http://schemas.
android.com/tools"
    android:id="@+id/main"
```

```
android:layout_width="match_
parent"
```

```
android:layout_height="match
_parent"
```

```
tools:context=".MainActivity
"
```

```
android:background="#005"
```

```
android:orientation="vertica
l"
```

```
android:gravity="center">
```

```
<TextView
```

```
android:layout_width="wrap_c
ontent"
```

```
android:layout_height="wrap_
content"
```

```
        android:text="Parul  
University"  
  
        android:textSize="34sp"  
  
        android:textColor="#FFF"  
  
        android:textStyle="bold"  
  
        tools:ignore="MissingConstraints"/>  
  
    </LinearLayout>
```

Activity_login

```
<?xml version="1.0"  
encoding="utf-8"?>  
    <LinearLayout  
        xmlns:android="http://schemas  
        .android.com/apk/res/androi  
        d"
```

```
xmlns:app="http://schemas.android.com/apk/res-auto"
```

```
xmlns:tools="http://schemas.android.com/tools"
```

```
    android:id="@+id/main"
```

```
    android:layout_width="match_parent"
```

```
    android:layout_height="match_parent"
```

```
    android:background="#FFC107"
```

```
    android:gravity="center"
```

```
    tools:context=".LoginActivity">
```

```
<Button
```

```
    android:id="@+id/btnLogin"
```

```
android:layout_width="wrap_content"
```

```
android:layout_height="wrap_content"
```

```
android:backgroundTint="#EFD  
EDE"
```

```
android:text="Login"
```

```
android:textColor="#454343"  
</>
```

```
</LinearLayout>
```

Activity_Home

```
<?xml version="1.0"  
encoding="utf-8"?>  
    <RelativeLayout  
xmlns:android="http://schemas  
.android.com/apk/res/androi  
d"
```

```
xmlns:app="http://schemas.android.com/apk/res-auto"
```

```
xmlns:tools="http://schemas.android.com/tools"
```

```
    android:id="@+id/main"
```

```
    android:layout_width="match_parent"
```

```
    android:layout_height="match_parent"
```

```
    android:background="#F1C02F"
```

```
    android:orientation="vertical"
```

```
    tools:context=".HomeActivity">
```

```
<Button
```

```
    android:id="@+id/btnLogout"
```

```
android:layout_width="wrap_c  
ontent"
```

```
android:layout_height="wrap_  
content"
```

```
android:layout_alignParentRi  
ght="true"
```

```
android:backgroundTint="#EFE  
6E6"
```

```
android:text="LOGOUT"
```

```
android:textColor="#292525"  
</>
```

```
</RelativeLayout>
```

MainActivity.java

```
package
com.example.sharedpreference;

import
android.content.Intent;
import
android.content.SharedPreference
ences;
import android.os.Bundle;
import
android.os.Handler;

import
androidx.activity.EdgeToEdge
;
import
androidx.appcompat.app.AppCo
mpatActivity;
import
androidx.core.graphics.Inset
s;
import
androidx.core.view.ViewCompa
t;
```

```
import
androidx.core.view.WindowIns
etsCompat;

    public class MainActivity
extends AppCompatActivity {

        @Override
        protected void
onCreate(Bundle
savedInstanceState) {

    super.onCreate(savedInstance
State);

    EdgeToEdge.enable(this);

    setContentView(R.layout.acti
vity_main);

        new
Handler().postDelayed(new
Runnable() {

            @Override
            public void
run() {
```

```

SharedPreferences pref=
getSharedPreferences("Login"
, MODE_PRIVATE);

Boolean
check=
pref.getBoolean("flag",
false);

Intent
iNext;

if (check)
{
iNext
= new
Intent (MainActivity.this,
HomeActivity.class);
} else {
iNext
= new
Intent (MainActivity.this,
LoginActivity.class);
}

startActivity(iNext);
}
}, 1000);

```

```
    }  
}
```

Login_activity.java

```
package  
com.example.sharedpreference;  
  
import  
android.content.Intent;  
import  
android.content.SharedPreference  
ences;  
import android.os.Bundle;  
import android.view.View;  
import  
android.widget.Button;  
  
import  
androidx.activity.EdgeToEdge  
;  
import  
androidx.appcompat.app.AppCo  
mpatActivity;
```

```
import  
androidx.core.graphics.Inset  
s;
```

```
import  
androidx.core.view.ViewCompa  
t;
```

```
import  
androidx.core.view.WindowIns  
etsCompat;
```

```
public class  
LoginActivity extends  
AppCompatActivity {  
    Button btnLogin;  
    @Override  
    protected void  
onCreate(Bundle  
savedInstanceState) {
```

```
super.onCreate(savedInstanceState);
```

```
EdgeToEdge.enable(this);
```

```
setContentView(R.layout.acti  
vity_login);
```

```
        btnLogin =  
findViewById(R.id.btnLogin);
```

```
btnLogin.setOnClickListener(  
new View.OnClickListener()  
    {  
        public void  
onClick(View view) {
```

```
    SharedPreferences pref =  
    getSharedPreferences("Login"  
    , MODE_PRIVATE);
```

```
    SharedPreferences.Editor  
editor= pref.edit();
```

```
editor.putBoolean("flag",  
true);
```

```
editor.apply();
```

```
Intent
```

```
iHome= new
```

```
Intent(LoginActivity.this,
HomeActivity.class);

startActivity(iHome);
    }
    });
}
}
```

Home_activity.java

```
package
com.example.sharedpreference;

import
android.content.SharedPrefer
ences;
    import android.os.Bundle;
    import android.view.View;
    import
android.widget.Button;
```

```
import
androidx.activity.EdgeToEdge
;

import
androidx.appcompat.app.AppCo
mpatActivity;

import
androidx.core.graphics.Inset
s;

import
androidx.core.view.ViewCompa
t;

import
androidx.core.view.WindowIns
etsCompat;


public class HomeActivity
extends AppCompatActivity {
    Button btnLogout;
    @Override
    protected void
onCreate(Bundle
savedInstanceState) {

super.onCreate(savedInstance
State);
```

```
EdgeToEdge.enable(this);

setContentView(R.layout.activity_home);

        btnLogout =
findViewById(R.id.btnLogout)
;

btnLogout.setOnClickListener
(new View.OnClickListener()
{
            @Override
            public void
onClick(View v) {

SharedPreferences pref=
getSharedPreferences("Login"
,  MODE_PRIVATE);

SharedPreferences.Editor
editor= pref.edit();
```

```

editor.putBoolean("flag",
false);

editor.apply();
        }
    });
}
}

```

Context Menu

mainActivity.xml

```

<?xml version="1.0"
encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas
s.android.com/apk/res/androi
d"

xmlns:app="http://schemas.an
droid.com/apk/res-auto"

```

```
xmlns:tools="http://schemas.  
android.com/tools"
```

```
    android:id="@+id/main"
```

```
    android:layout_width="match_  
parent"
```

```
    android:layout_height="match_  
parent"
```

```
    tools:context=".MainActivity  
"
```

```
    android:orientation="vertica  
l"
```

```
        android:gravity="center">
```

```
    <TextView
```

```
    android:layout_width="wrap_c  
ontent"
```

```
    android:layout_height="wrap_  
content"
```

```
        android:text="Long
```

```
Press Me"
```

```
        android:textSize="30sp"

        android:textStyle="bold"
            android:id="@+id/tv"
        />
</LinearLayout>
```

MainActivity.java.....

```
package
com.example.contextmenu;

import
android.graphics.Color;
import android.os.Bundle;
import
android.os.PersistableBundle
;
import
android.view.ContextMenu;
import
android.view.MenuItem;
import android.view.View;
```

```
import
android.widget.LinearLayout;
import
android.widget.TextView;
```

```
import
androidx.activity.EdgeToEdge
;
import
androidx.annotation.NonNull;
import
androidx.annotation.Nullable
;
import
androidx.appcompat.app.AppCompatActivity;
import
androidx.core.graphics.Insets;
import
androidx.core.view.ViewCompat;
import
androidx.core.view.WindowInsetsCompat;
```

```
public class MainActivity
extends AppCompatActivity {
    TextView tv;
    LinearLayout main;
    @Override
    protected void
onCreate(Bundle
savedInstanceState) {

    super.onCreate(savedInstanceState);

    EdgeToEdge.enable(this);

    setContentView(R.layout.activity_main);

    tv=
findViewById(R.id.tv);
    main =
findViewById(R.id.main);

    registerForContextMenu(tv);
}
```

```
        @Override
        public void
        onCreateContextMenu(ContextM
        enu menu, View v,
        ContextMenu.ContextMenuInfo
        menuInfo) {

            super.onCreateContextMenu(me
            nu, v, menuInfo);

            menu.setHeaderTitle("Choose
            a Color");

            menu.add(0,
            v.getId(), 0, "Light
            Yellow");

            menu.add(0,
            v.getId(), 0, "purple");

            menu.add(0,
            v.getId(), 0, "green");

        }
```

```
        @Override
        public boolean
        onContextItemSelected(@NonNu
        ll MenuItem item) {
```

```
        if (item.getTitle()
== "Light Yellow") {

main.setBackgroundColor(Color
r.argb(100, 225, 193, 7));
        } if
(item.getTitle() == "green")
{

main.setBackgroundColor(Color
r.argb(100, 3, 209, 111));
        } if
(item.getTitle() ==
"purple") {

main.setBackgroundColor(Color
r.argb(100, 89, 64, 187));
        }
        return true;
    }
}
```